

FinX: Data Ingestion (Module 1) Demo Procedure

Here is your exact checklist to start everything from a completely closed, blank state.

Step 1: Wake up Docker (Crucial First Step)

Since you closed everything, your background services (like Kafka) are turned off.

1. Open the Docker Desktop application on your computer.
2. Wait a minute for the icon in the bottom-left corner to turn green and say "Engine running".
(Do not skip this, or the script will crash with a Timeout error).

Step 2: Open your Project

1. Open your terminal or VS Code.
2. Make sure you are in your project folder (FinX).

Step 3: Turn on the Streaming Infrastructure

1. In your terminal, start the Kafka cluster by running:
`docker-compose up -d`
2. Wait a few seconds for it to say 'Started' for both zookeeper and kafka.

Step 4: Run the Demonstration

Now your system is ready for the presentation.

To demonstrate the "Huge" Dataset (the fast stream):

```
python streaming/dataset_replay.py --dataset huge
```

(Press Ctrl + C when you are ready to stop it).

To demonstrate the "Controlled" Dataset (slow stream with seeded anomaly):

```
python streaming/dataset_replay.py --dataset controlled
```

Step 5: Shutting Down (After the demo)

When you are completely finished with your presentation and want to close everything down cleanly, run:

```
docker-compose down
```